Signals and Systems
Assignment-6: Solution

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Exercise 1 (1)

Find the causal signal whose Laplace transform is given by:
$X(s) = \frac{2s^2+7s+9}{(s+2)^2}$

- Since $X(s)$ is nonproper:

$$X(s) = \frac{2(s^2+4s+4)-s+1}{(s+2)^2} = 2 + \frac{-s+1}{(s+2)^2} \triangleq 2 + X_1(s)$$

- where:

$$X_1(s) = \frac{-s+1}{(s+2)^2} = \frac{r_1}{(s+2)^2} + \frac{r_2}{s+2}$$
$$r_1 = [-s+1]_{s=-2} = 3$$

- Let $s = 1$:

$$0 = \frac{r_1}{9} + \frac{r_2}{3} \Rightarrow r_2 = -\frac{r_1}{3} = -1$$

Exercise 1 (2)

- Consequently:

$$X(s) = 2 + \frac{3}{(s+2)^2} - \frac{1}{s+2}$$

- Finally:

$$x(t) = 2\delta(t) + \left[3te^{-2t} - e^{-2t}\right] u(t)$$

Exercise 2 (1)

The Laplace transform of a signal $x(t)$ is given by: $X(s) = \frac{4s}{s^2+11s+24}e^{-2s}$. Using the initial value theorem and the final value theorem evaluate $x(0^+)$ and $\lim_{t\to+\infty} x(t)$.

- Initial value theorem:

$$x(0^+) = \lim_{s\to+\infty} sX(s) = \lim_{s\to+\infty} \frac{4s^2}{s^2+11s+24}e^{-2s} = 0$$

- Final value theorem:

$$\lim_{t\to+\infty} x(t) = \lim_{s\to 0} sX(s) = \lim_{s\to 0} \frac{4s^2}{s^2+11s+24}e^{-2s} = 0$$

Exercise 2 (2)

- X_s can be written as:

$$X(s) = \frac{4s}{s^2 + 11s + 24} e^{-2s} = X_1(s) e^{-2s}$$

- Applying the partial fraction expansion on $X_1(s)$:

$$X_1(s) = \frac{4s}{s^2 + 11s + 24} = \frac{4s}{(s+3)(s+8)} = \frac{r_1}{s+3} + \frac{r_2}{s+8}$$

$$r_1 = \left[\frac{4s}{s+8}\right]_{s=-3} = -\frac{12}{5} \; ; r_2 = \left[\frac{4s}{s+3}\right]_{s=-8} = \frac{32}{5}$$

- Consequently:

$$X_1(s) = -\frac{12}{5}\frac{1}{s+3} + \frac{32}{5}\frac{1}{s+8}$$

- Therefore:

$$x_1(t) = \left[-\frac{12}{5}e^{-3t} + \frac{32}{5}e^{-8t}\right] u(t)$$

Exercise 2 (3)

- Given that: $\mathcal{L}(x(t-t_0)) = X(s)e^{-st_0}$:

$$\begin{aligned} & X(s) = X_1(s)e^{-2s} \\ \Rightarrow\ & x(t) = x_1(t-2) \\ \Rightarrow\ & x(t) = \left[-\frac{12}{5}e^{-3(t-2)} + \frac{32}{5}e^{-8(t-2)}\right]u(t-2) \end{aligned}$$

- $x(0^+)$ can be calculated from:

$$x(0^+) = \left[-\frac{12}{5}e^{6} + \frac{32}{5}e^{16}\right]u(-2) = 0$$

since $u(-2) = 0$.

- $\lim_{t\to+\infty} x(t)$ can be calculated from:

$$\lim_{t\to+\infty} x(t) = \lim_{t\to+\infty}\left[-\frac{12}{5}e^{-3(t-2)} + \frac{32}{5}e^{-8(t-2)}\right]u(t-2) = 0$$

Exercise 3 (1)

Find the Laplace transform of $x(t) = t^2 e^{-at}\delta(t-2)$.

- $x(t)$ can be written as:

$$x(t) = t^2 e^{-at}\delta(t-2) = 4e^{-2a}\delta(t-2)$$

- Consequently:

$$X(s) = \mathcal{L}(x(t)) = 4e^{-2a}e^{-2s} = 4e^{-2(a+s)}$$

Exercise 3 (2)

Find the Laplace transform of $x(t) = te^{-(t-a)}u(t-a)$.

- $x(t)$ can be written as:

$$x(t) = te^{-(t-a)}u(t-a) = (t-a)e^{-(t-a)}u(t-a) + ae^{-(t-a)}u(t-a)$$

- Given that:

$$\mathcal{L}[e^{-t}u(t)] = \frac{1}{s+1} \Rightarrow \mathcal{L}[e^{-(t-a)}u(t-a)] = \frac{1}{s+1}e^{-as}$$

$$\mathcal{L}[te^{-t}u(t)] = \frac{1}{(s+1)^2} \Rightarrow \mathcal{L}[(t-a)e^{-(t-a)}u(t-a)] = \frac{1}{(s+1)^2}e^{-as}$$

- Consequently:

$$\begin{aligned}\mathcal{L}[x(t)] &= \frac{1}{(s+1)^2}e^{-as} + a\frac{1}{s+1}e^{-as} \\ &= \frac{1}{(s+1)^2}\left[1 + a(s+1)\right]e^{-as}\end{aligned}$$

Exercise 3 (3)

Second Method:

- Let $f(t) = e^{-(t-a)}u(t-a)$ then:

$$F(s) = \frac{1}{s+1}e^{-as}$$

- Applying the complex-frequency differentiation property, then $\mathcal{L}[x(t)] = \mathcal{L}[tf(t)] = -\frac{\mathrm{d}F(s)}{\mathrm{d}s}$. Consequently:

$$\begin{aligned} X(s) &= -\frac{\mathrm{d}}{\mathrm{d}s}\left[\frac{1}{s+1}e^{-as}\right] \\ &= -\left[-\frac{1}{(s+1)^2}e^{-as} - \frac{a}{s+1}e^{-as}\right] \\ &= \frac{1}{(s+1)^2}\left[1 + a(s+1)\right]e^{-as} \end{aligned}$$

Exercise 3 (4)

Find the Laplace transform of $x(t) = e^{-t}u(t-a)$.

- $x(t)$ can be written as:

$$x(t) = e^{-t}u(t-a) = e^{-a}e^{-(t-a)}u(t-a)$$

- Therefore:

$$\begin{aligned} X(s) = \mathcal{L}[x(t)] &= e^{-a}\left[\frac{1}{s+1}e^{-as}\right] \\ &= \frac{1}{s+1}e^{-a(s+1)} \end{aligned}$$

Exercise 3 (5)

Find the Laplace transform of $x(t) = t^2 u(t-3)$.

- $x(t)$ can be written in terms of $t-3$ as:

$$\begin{aligned} x(t) &= \left[(t-3)^2 + 6t - 9\right] u(t-3) \\ &= \left[(t-3)^2 + 6(t-3+3) - 9\right] u(t-3) \\ &= \left[(t-3)^2 + 6(t-3) + 9\right] u(t-3) \end{aligned}$$

- Consequently:

$$X(s) = \frac{2}{s^3} e^{-3s} + \frac{6}{s^2} e^{-3s} + \frac{9}{s} e^{-3s}$$

Exercise 3 (6)

Find the Laplace transform of $x(t) = 5\cos(2t-1)u(t)$.

- $x(t)$ can be written as:

$$\begin{aligned} x(t) &= 5\cos(2t-1)u(t) \\ &= 5\left[\cos(2t)\cos 1 + \sin(2t)\sin 1\right]u(t) \\ &= 5\cos 1\cos(2t)u(t) + 5\sin 1\sin(2t)u(t) \end{aligned}$$

- Consequently:

$$\begin{aligned} X(s) &= 5\cos 1\frac{s}{s^2+(2)^2} + 5\sin 1\frac{2}{s^2+(2)^2} \\ &= \frac{5}{s^2+4}\left[\cos(1)s + 2\sin 1\right] \end{aligned}$$

Exercise 3 (7)

Find the Laplace transform of $x(t) = \sin(3t - 1)u(t - 1)$.

- $x(t)$ can be written as:

$$\begin{aligned} x(t) &= \sin(3(t-1)+2)u(t-1) \\ &= \cos(2)\sin(3(t-1))u(t-1) + \sin(2)\cos(3(t-1))u(t-1) \end{aligned}$$

- Consequently:

$$X(s) = \cos(2)\frac{3}{s^2+9}e^{-s} + \sin(2)\frac{s}{s^2+9}e^{-s}$$

Exercise 4 (1)

- Taking the Laplace transform of both sides of the DE implies that:

$$\left[s^2X(s) - sx(0^-) - x'(0^-)\right] - 2\left[sX(s) - x(0^-)\right] + 4X(s) = \frac{1}{s}$$

$$\Rightarrow \quad \left[s^2 - 2s + 4\right]X(s) = \frac{1}{s} + 4 = \frac{4s+1}{s}$$

$$\Rightarrow \quad X(s) = \frac{4s+1}{s\left(s^2 - 2s + 4\right)} = \frac{4s+1}{s\left((s-1)^2 + 3\right)}$$

- The PFE of $X(s)$ is given by:

$$X(s) = \frac{r_1}{s} + \frac{r_2 s + r_3}{(s-1)^2 + 3}$$

where:

$$r_1 = \left[\frac{4s+1}{(s-1)^2 + 3}\right]_{s=0} = \frac{1}{4}$$

Exercise 4 (2)

- r_2 and r_3 can be determined by taking two particular values of s such as $s = 1$ and $s = -1$:

$$s = 1 \quad \Rightarrow \quad \frac{5}{3} = \frac{1}{4} + \frac{r_2 + r_3}{3} \quad \Rightarrow \quad r_2 + r_3 = \frac{17}{4}$$

$$s = -1 \quad \Rightarrow \quad \frac{3}{7} = -\frac{1}{4} + \frac{-r_2 + r_3}{7} \quad \Rightarrow \quad -r_2 + r_3 = \frac{19}{4}$$

- Solving the above equations results in:

$$r_2 = -\frac{1}{4} \;\; ; \;\; r_3 = \frac{9}{2}$$

implying that:

$$X(s) = \frac{1}{4}\left[\frac{1}{s} - \frac{s}{(s-1)^2 + 3} + 18\frac{1}{(s-1)^2 + 3}\right]$$

$$= \frac{1}{4}\left[\frac{1}{s} - \frac{s-1}{(s-1)^2 + 3} + \frac{17}{\sqrt{3}}\frac{\sqrt{3}}{(s-1)^2 + 3}\right]$$

Exercise 4 (3)

- Given that:

$$\mathcal{L}\left[e^{\alpha t}\cos(\omega_0 t)u(t)\right] = \frac{s-\alpha}{(s-\alpha)^2+\omega_0^2}$$
$$\mathcal{L}\left[e^{\alpha t}\sin(\omega_0 t)u(t)\right] = \frac{\omega_0}{(s-\alpha)^2+\omega_0^2}$$

then $x(t)$ can be written as:

$$x(t) = \frac{1}{4}\left[1 - e^t\cos(\sqrt{3}t) + \frac{17}{\sqrt{3}}e^t\sin(\sqrt{3}t)\right]u(t)$$

Exercise 5 (1)

Find the inverse Laplace transform of $X(s) = \frac{20}{s^2+4s+3}$:

- $X(s)$ can be written as:

$$X(s) = \frac{20}{(s+1)(s+3)} = \frac{r_1}{s+1} + \frac{r_2}{s+3}$$

  where applying the PFE: $r_1 = 10$ and $r_2 = -10$.

- Consequently, $x(t)$ can be written as:

$$x(t) = 10\left[e^{-t} - e^{-3t}\right]u(t)$$

Exercise 5 (2)

Find the inverse Laplace transform of $X(s) = \frac{s}{s^2+4s+4}$:

- $X(s)$ can be written as:

$$\begin{aligned} X(s) &= \frac{s}{s^2+4s+4} = \frac{s}{(s+2)^2} \\ &= \frac{s+2}{(s+2)^2} - \frac{2}{(s+2)^2} \\ &= \frac{1}{s+2} - 2\frac{1}{(s+2)^2} \end{aligned}$$

implying that:

$$x(t) = \left[e^{-2t} - 2te^{-2t}\right] u(t) = e^{-2t}\left[1-2t\right] u(t)$$

Exercise 5 (3)

Find the inverse Laplace transform of $X(s) = \frac{2s}{s^2+2s+13}$:

- $X(s)$ can be written as:

$$
\begin{aligned}
X(s) &= \frac{2s}{s^2+2s+13} = \frac{2s}{(s+1)^2+12} \\
&= 2\left[\frac{s+1}{(s+1)^2+12} - \frac{1}{(s+1)^2+12}\right] \\
&= 2\left[\frac{s+1}{(s+1)^2+(\sqrt{12})^2} - \frac{1}{\sqrt{12}}\frac{\sqrt{12}}{(s+1)^2+(\sqrt{12})^2}\right]
\end{aligned}
$$

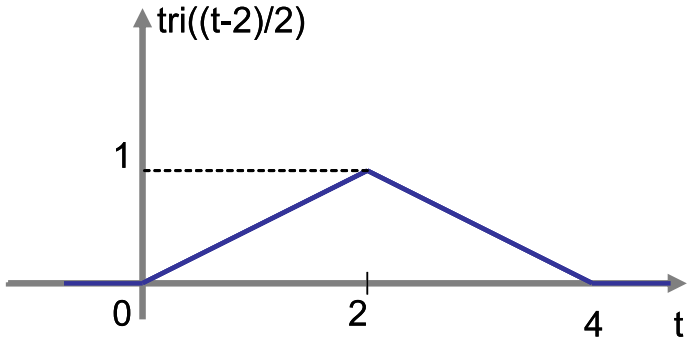
implying that:

$$
\begin{aligned}
x(t) &= 2\left[e^{-t}\cos(\sqrt{12}t) - \frac{1}{\sqrt{12}}e^{-t}\sin(\sqrt{12}t)\right]u(t) \\
&= 2e^{-t}\left[\cos(\sqrt{12}t) - \frac{1}{\sqrt{12}}\sin(\sqrt{12}t)\right]u(t)
\end{aligned}
$$

Exercise 6 (1)

Let $x(t) = \mathrm{tri}\left(\frac{t-2}{2}\right)$. Calculate a_1, a_2, b_1, b_2 and c_1, c_2 such that:

$$x(t) = (a_1 t + a_2)u(t) + (b_1 t + b_2)u(t-2) + (c_1 t + c_2)u(t-4)$$



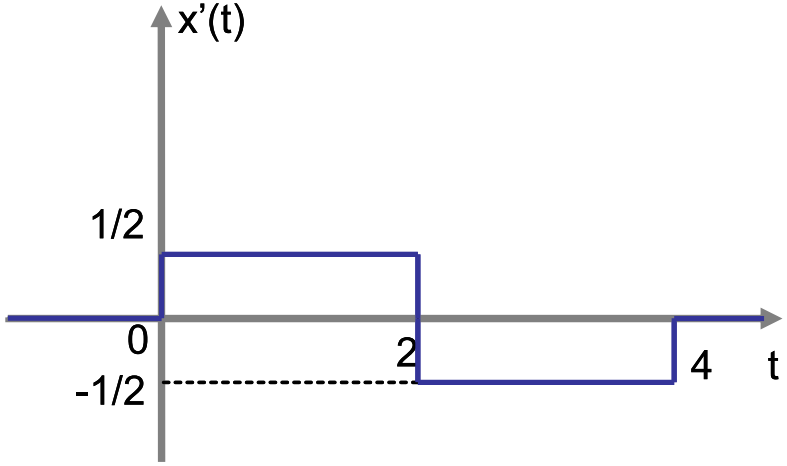
- $x(t)$ can be written as:

$$\begin{aligned} x(t) &= \frac{1}{2}\mathrm{ramp}(t) - \mathrm{ramp}(t-2) + \frac{1}{2}\mathrm{ramp}(t-4) \\ &= \frac{1}{2}tu(t) - (t-2)u(t-2) + \frac{1}{2}(t-4)u(t-4) \end{aligned}$$

- $a_1 = \frac{1}{2}$, $a_2 = 0$, $b_1 = -1$, $b_2 = 2$, $c_1 = \frac{1}{2}$ and $c_2 = -2$.
- Therefore:

$$X(s) = \frac{1}{2}\frac{1}{s^2} - \frac{1}{s^2}e^{-2s} + \frac{1}{2}\frac{1}{s^2}e^{-4s}$$

Exercise 6 (2)



- $x'(t)$ can be written as:

$$x'(t) = \frac{1}{2}u(t) - u(t-2) + \frac{1}{2}u(t-4)$$

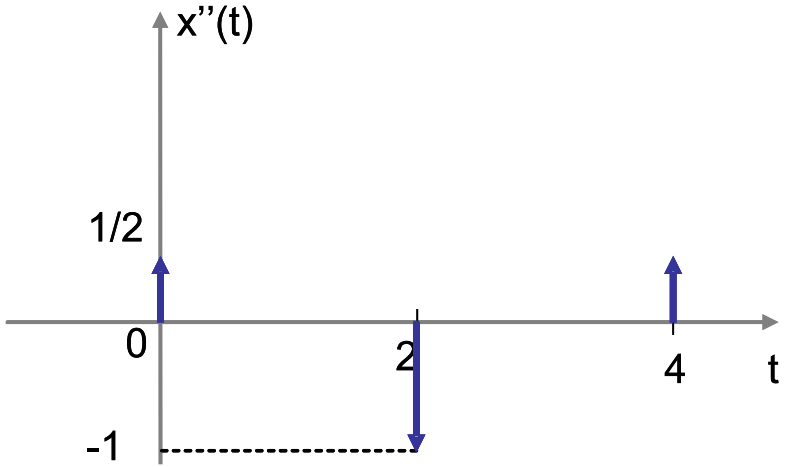
- Consequently:

$$X'(s) = \frac{1}{2}\frac{1}{s} - \frac{1}{s}e^{-2s} + \frac{1}{2}\frac{1}{s}e^{-4s}$$

- Therefore:

$$X(s) = \frac{X'(s)}{s} = \frac{1}{2}\frac{1}{s^2} - \frac{1}{s^2}e^{-2s} + \frac{1}{2}\frac{1}{s^2}e^{-4s}$$

Exercise 6 (3)



- $x''(t)$ can be written as:

$$x''(t) = \frac{1}{2}\delta(t) - \delta(t-2) + \frac{1}{2}\delta(t-4)$$

- Consequently:

$$X''(s) = \frac{1}{2} - e^{-2s} + \frac{1}{2}e^{-4s}$$

- Therefore:

$$X(s) = \frac{X''(s)}{s^2} = \frac{1}{2}\frac{1}{s^2} - \frac{1}{s^2}e^{-2s} + \frac{1}{2}\frac{1}{s^2}e^{-4s}$$

Exercise 7

- From the block diagram of the system:

$$y(t) = \int_0^t [x(\tau) - 4y(\tau)]\mathrm{d}\tau$$

- Taking the Laplace transform of both sides of this equation:

$$Y(s) = \frac{X(s) - 4Y(s)}{s} \quad \Rightarrow \quad H(s) = \frac{Y(s)}{X(s)} = \frac{1}{s+4}$$

- For $x(t) = u(t)$, $X(s) = \frac{1}{s}$ resulting in:

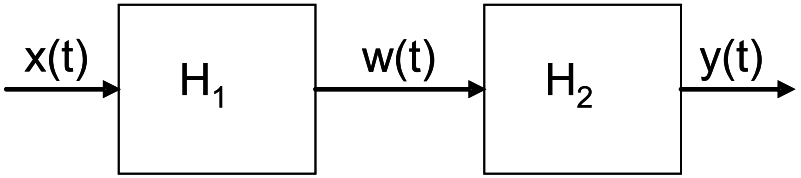
$$Y(s) = \frac{1}{s(s+4)} = \frac{r_1}{s} + \frac{r_2}{s+4}$$

  where $r_1 = \frac{1}{4}$ and $r_2 = -\frac{1}{4}$ using the PFE.
- Consequently:

$$y(t) = \frac{1}{4}\left[1 - e^{-4t}\right]u(t)$$

Exercise 8 (1)

Determine the transfer function of the system: $H(s) = \frac{Y(s)}{X(s)}$.



- The impulse response of H_1 is given by $h_1(t) = e^{-2t}u(t)$, then the transfer function is given by:

$$H_1(s) = \mathcal{L}[h_1(t)] = \frac{1}{s+2}$$

- H_2 is described by the differential equation:

$$\frac{\mathrm{d}^2 y(t)}{\mathrm{d}t^2} + 3\frac{\mathrm{d}y(t)}{\mathrm{d}t} = \frac{\mathrm{d}w(t)}{\mathrm{d}t} - w(t)$$

$$\Rightarrow \left[s^2 Y(s) - sy(0^-) - y'(0^-)\right] + 3\left[sY(s) - y(0^-)\right] = \left[sW(s) - w(0^-)\right] - W(s)$$

- Given that the system was initially at rest:

$$y(0^-) = y'(0^-) = w(0^-) = 0$$

Exercise 8 (2)

- Consequently:

$$s^2Y(s) + 3sY(s) = sW(s) - W(s)$$
$$\Rightarrow H_2(s) = \frac{Y(s)}{W(s)} = \frac{s-1}{s^2+3s}$$

- Therefore, the transfer function of the system is given by:

$$H(s) = \frac{Y(s)}{X(s)} = H_1(s)H_2(s) = \frac{s-1}{s(s+2)(s+3)}$$

Exercise 8 (3)

Determine whether the above system is stable or not.

- The poles of $H(s)$ are $p_1 = 0$, $p_2 = -2$ and $p_3 = -3$.
- Given that p_1 is not in the open LHP, then the system is not stable.
  - Note that for the system to be stable, <u>all</u> poles must lie in the open LHP.

Exercise 8 (4)

Write down the differential equation relating $y(t)$ to $x(t)$.

- The differential equation between $x(t)$ and $y(t)$ can be obtained from:

$$\left(s^3 + 5s^2 + 6s\right) Y(s) = (s-1)X(s)$$

- Implying that:

$$\frac{\mathrm{d}^3 y(t)}{\mathrm{d}t^3} + 5\frac{\mathrm{d}^2 y(t)}{\mathrm{d}t^2} + 6\frac{\mathrm{d}y(t)}{\mathrm{d}t} = \frac{\mathrm{d}x(t)}{\mathrm{d}t} - x(t)$$

Exercise 8 (5)

Determine the response of the system when $x(t) = e^{-3t}u(t)$.

- The Laplace transform of the output is given by:

$$Y(s) = X(s)H(s) = \frac{1}{(s+3)}H(s) = \frac{s-1}{s(s+2)(s+3)^2}$$

- Applying the partial fraction expansion on $Y(s)$:

$$\begin{aligned} Y(s) &= \frac{r_1}{s} + \frac{r_2}{s+2} + \frac{r_3}{(s+3)^2} + \frac{r_4}{s+3} \\ &= -\frac{1}{18}\frac{1}{s} + \frac{3}{2}\frac{1}{s+2} - \frac{4}{3}\frac{1}{(s+3)^2} - \frac{13}{9}\frac{1}{s+3} \end{aligned}$$

- Therefore:

$$y(t) = \left[-\frac{1}{18} + \frac{3}{2}e^{-2t} - \frac{4}{3}te^{-3t} - \frac{13}{9}e^{-3t}\right]u(t)$$

Exercise 8 (6)

Determine the response of the system when $x(t) = e^{-3t}\text{rect}\left(\frac{t-1}{2}\right)$.

- $x(t)$ can be written as:

$$\begin{aligned} x(t) &= e^{-3t}\left[u(t) - u(t-2)\right] \\ &= e^{-3t}u(t) - e^{-6}e^{-3(t-2)}u(t-2) \end{aligned}$$

- Since the system is LTI, then from the previous part:

$$\begin{aligned} y(t) = &\left[-\frac{1}{18} + \frac{3}{2}e^{-2t} - \frac{4}{3}te^{-3t} - \frac{13}{9}e^{-3t}\right]u(t) \\ &-e^{-6}\left[-\frac{1}{18} + \frac{3}{2}e^{-2(t-2)} - \frac{4}{3}(t-2)e^{-3(t-2)} - \frac{13}{9}e^{-3(t-2)}\right]u(t-2) \end{aligned}$$